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Section: C

Course: Computer Networks

Assignment: 02

Question:

You have already learnt terms like GBN, SR. Let us consider a network scenario which is with substantially high error rates and often faces long propagation delays when sending packets. Which of the two mechanisms would be better in such a scenario and why? Explain and justify your answer.

Answer:

In a network with high error rates and long propagation delays, Selective Repeat (SR) is better than Go-Back-N (GBN).

In GBN, if one packet is lost, all packets after that packet are retransmitted, even if they were received correctly. This wastes bandwidth and increases delay.

In SR, only the lost or corrupted packet is retransmitted. The receiver can also buffer correctly received packets. Therefore, SR provides better bandwidth utilization and higher efficiency.

So, Selective Repeat is more suitable for networks with high error rates and long propagation delays.